

DEEP STRUCTURE

MASTERING STUDIO

Guide

All functions and how to use them, explained step by step

Version 1.17.0 · November 2026

deepstructure.world/studio

WELCOME

What the studio does

The Deep Structure Mastering Studio turns a finished song (for example a WAV from Suno) into a master ready for release. It measures the song, shows weak spots, repairs typical AI artefacts (including steady whistling and noise with “Clean Song”) and drums, cuts and exports as WAV, OGG or MP3. With **My Style** it learns from your confirmed masters, and in the **Album** tab you check and approve whole albums. The interface is available in English and German.

- **Everything runs on your device.** No upload, no cloud, no account. Your audio file never leaves the browser.
- **The original stays unchanged.** The studio works on a copy; only what you export is saved.
- **Nothing happens automatically.** Suggestions and the assistant are starting points. Your ears decide how it sounds in the end.

Where the studio runs

Version	How to start it
Website	Open **deepstructure.world/studio** in the browser. After the first visit it also works offline. On an Android phone you can add it to the home screen like an app via the browser menu (“Add to Home screen”).
PC (Windows)	In the studio folder, double-click **Desktop-Verknuepfung-erstellen.vbs** once; the studio then sits on the desktop as an icon. Alternatively start **Studio starten.vbs** directly. It opens in its own window.
In Claude	As a page in the Claude app. Only there does AI mastering run via your Claude account (see chapter 5).

Formats you can load: WAV (8/16/24/32 bit, also 32/64-bit float, also multichannel), MP3, FLAC, M4A/AAC and OGG. The file is loaded at its original sample rate.

Contents

1 Quick start in 8 steps 3
2 The interface at a glance 4
3 Header and playback 6
4 Waveform and cutting 7
5 Song check, AI artefacts, Clean Song 9
6 Equalizer 14
7 Dynamics, saturation, stereo 16
8 Limiter and loudness 18
9 Drum repair 19
10 Range, reference, analysis 21
11 Game data 26
12 Meters 27
13 Exporting and saving 28
14 My Style 29
15 Album: review and approve 30
16 MIDI controller 33
17 Keys and gestures 35
18 Tips and help 36

1 Quick start in 8 steps

The shortest way from a finished song to a master. Each step is explained in more detail in the later chapters.

1. **Open a song**
   Tap **Open song** at the top left or simply drag the file into the window. The studio measures the song right away.

2. **Read the song check**
   The **Song check** tab shows a guide value and concrete notes (tonal balance, stereo, AI artefacts). Suggestions with a button can be applied with one click.

3. **Press Assistant**
   Right next to the tabs. It analyses the song and sets a cautious starting point. A technically clean song stays almost unchanged.

4. **Compare**
   Play (`Space`) and switch between **Master** and **Original** at the top (`M` / `O`). Leave **Level match** on, so you compare the sound and not the volume.

5. **Fine-tune by ear**
   Adjust in the Equalizer, Dynamics, Saturation and Stereo tabs. The dot on a tab shows whether the module is switched on.

6. **Set loudness**
   Limiter tab: choose a loudness target (Spotify/YouTube −14 LUFS) and press **Set drive to target**. Leave the ceiling at −1 dB.

7. **Check**
   The **Analysis** tab shows the exact values for the whole song, original and master side by side.

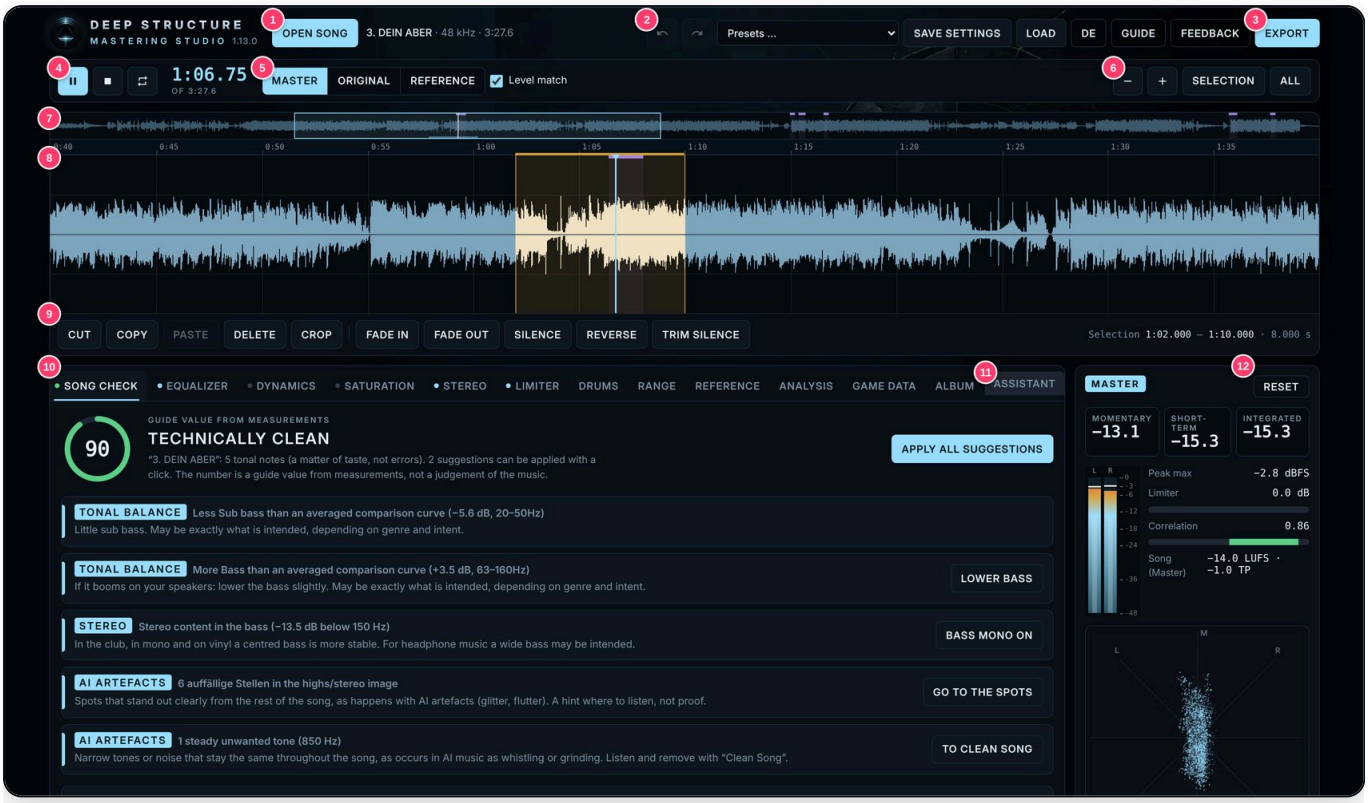
8. **Export**
   Export at the top right: choose a format (WAV 24 bit for distribution, OGG for games, MP3 for sharing) and **Start export**. The file goes to your downloads folder.

> **For Suno songs**
> If the drums sound sandy, metallic or chopped off, use the **Drums** tab before step 5 (chapter 9). For glittering highs, fluttering stereo, steady whistling or noise there is the **AI artefacts** card with **Clean Song** in the song check.

> **Save your cuts**
> Edits in the waveform are not saved automatically. If you have only cut, save that version with Export → Content **Edit only** before closing the window.

2 The interface at a glance

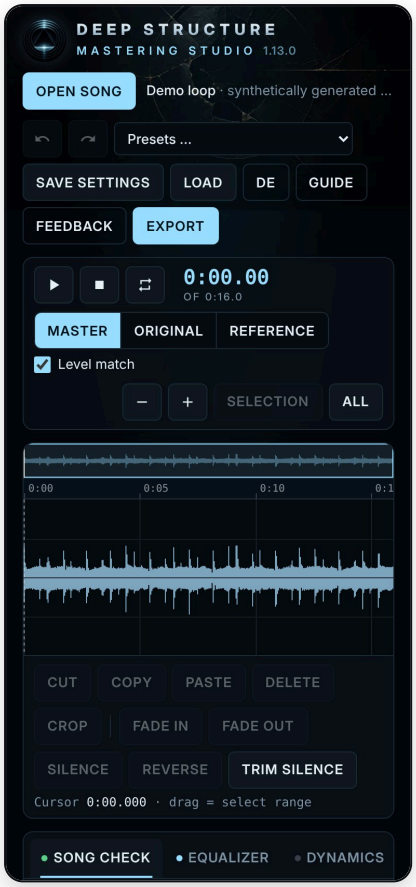
From top to bottom: header, playback, waveform with cutting tools, below it the modules in tabs and on the right the meters.



The studio on a PC with a song loaded, a range marked and playback running

1	**Open song** · next to it the name, sample rate and length of the loaded song
2	**Undo / redo, presets, save and load settings, language (EN / DE), guide, feedback**
3	**Export** · save the finished file
4	**Play, stop, loop** · next to it the time display
5	**Monitoring:** master, original or reference, and **level match**
6	**Zoom:** out, in, to the selection, whole song
7	**Song overview:** the whole song; the bright frame shows the visible section
8	**Waveform** with timeline, playback position and selection (yellow)
9	**Cutting tools** · on the right start, end and length of the selection
10	**Module tabs** · a dot shows that the module is switched on
11	**Assistant** · sets a starting point for the song · next to it **My Style** (chapter 14)
12	**Meters** · loudness, peaks, limiter, stereo image (chapter 12)

On the phone



Same functions, arranged one below the other

On the phone all sections are stacked. You operate everything with your fingers:

- **Tap** in the waveform to set the playback position.
- **Drag** to mark a range.
- **Two fingers** apart or together to zoom.
- Dragging in the timeline at the top of the waveform moves the playback position.
- In the equalizer, **two fingers** on a point change the bandwidth (Q).

Use headphones or good speakers. Phone speakers hardly reproduce bass and stereo image.

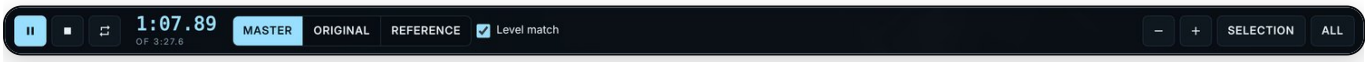
3 Header and playback



Button	Function
Open song	Load an audio file. Drag and drop into the window works the same way.
↶ / ↷	Undo and redo for all edits to the song (cuts, smoothing, drum repair, range tools). Hovering shows which step will be undone.
Presets	Ready-made settings as a starting point: *Neutral (nothing changed)*, *Streaming clear (−14 LUFS)*, *Loud & punchy (−9 LUFS)*, *Warm & analogue*, *Vinyl premaster*, *Suno mix starting point*. A preset also sets the matching loudness target. Then press “Set drive to target” in the limiter.
Save settings	Save all control positions as a file (*.dsmaster.json*), for example one per song.
Load	Load a saved settings file. If the file is invalid, everything stays as it was.
EN / DE	Switch the interface language (English / German). The button shows the other language. The choice is remembered on the device; on first start it follows the browser language. Measurements, files and settings stay unchanged.
Guide	Download this guide as a PDF (with the German interface, the German version). In the PC version it opens in the PDF viewer, where you can save or print it.
Feedback	Write feedback to Deep Structure (chapter 13).
Export	Save the finished master or the cut version (chapter 13).

The browser remembers your settings automatically, also across updates.

Playback and monitoring



▶ / ❚❚	Play and pause (`Space`). Playback starts at the playback position.
■	Stop, back to the start point.
Loop	Plays the marked selection over and over (`L`). Ideal for comparing one spot.
Master	You hear the song with all switched-on modules (`M`).
Original	You hear the loaded file without mastering (`0`). If you have edited the song but not changed its length, the meter panel says “Original file”. After a cut it says “Without mastering”: then you hear your cut version without modules.
Reference	You hear a loaded comparison song (`R`, chapter 10).
Level match	Brings all three sources to the same loudness. This way you compare the sound, not the volume. Otherwise louder always sounds “better”.
− / + / Selection / All	Zoom out, zoom in (`−` / `+`), zoom to the marked selection, show the whole song.

4 Waveform and cutting



At the top the song overview, below it two tracks: master and original, with the selection marked in yellow; at the bottom the cutting tools

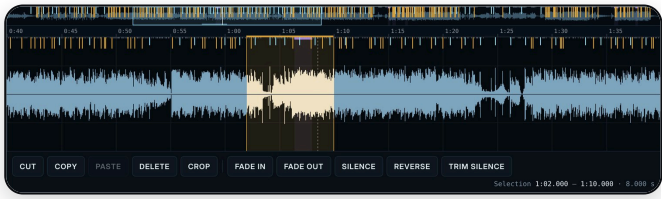
Two tracks: At the top is the **master**, as it sounds after the whole mastering chain (equalizer, compressor, limiter). It is recalculated shortly after every change; until then it says “updating”. At the bottom is the **original**, the loaded file, which never changes. The track you are hearing is bright and marked with ▶. **Clicking into a track switches monitoring to it**, like the Master / Original buttons or the M and O keys, without interrupting playback. With “Level match” you hear both at the same loudness. Cuts and repairs act on the master; after cuts the lower track shows “Without mastering”, because the file no longer lines up in time.

Using the mouse

- **Click:** set the playback position
- **Drag:** mark a range
- **Drag the edge of the selection:** make the selection smaller or larger
- **Shift + click:** extend the selection up to this point
- **Timeline at the top:** click or drag to move the playback position
- **Mouse wheel:** zoom, **Shift + mouse wheel:** scroll sideways
- **Double-click:** show the whole song
- **Song overview:** click or drag to jump to that spot
- `Esc` clears the selection, `Ctrl`+`A` selects everything

Coloured marks

Violet marks above the waveform show suspicious spots from the AI artefact search. After “Find hits” in the Drums tab, marks for the detected hits appear as well: gold = kick, light blue = snare, grey = hi-hat. Comments from the album appear as marks too.



Waveform with drum marks

The cutting tools

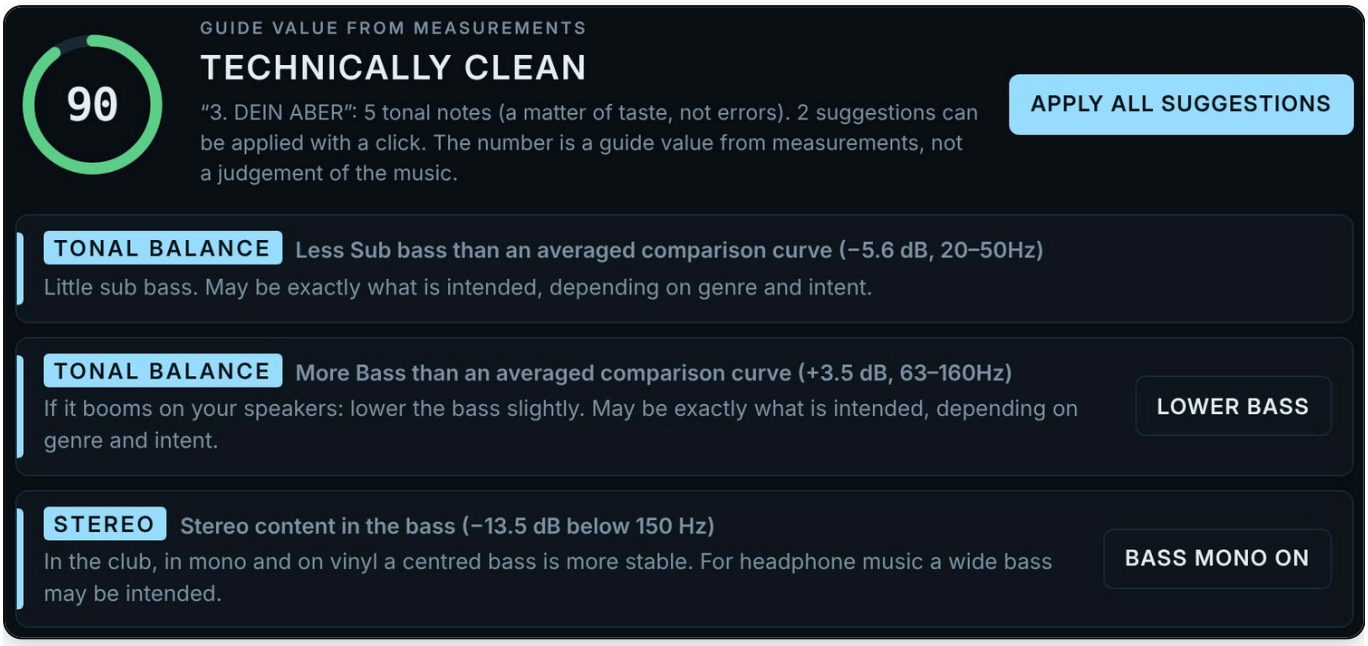
Button	Function	Key
Cut	Remove the selection and keep it in memory	`Ctrl`+`X`
Copy	Keep the selection in memory	`Ctrl`+`C`
Paste	Insert the stored piece at the playback position or in place of the selection	`Ctrl`+`V`
Delete	Remove the selection	`Del`
Crop	Keep only the selection, remove the rest	

Fade in / fade out	Fade in or out over the length of the selection
Silence	Mute the selection
Reverse	Play the selection backwards
Trim silence	Remove silence at the start and end of the song

Everything can be undone: `Ctrl`+`Z` takes back every step, `Ctrl`+`Y` restores it.

5 Song check, AI artefacts, Clean Song

When a song loads, the studio checks loudness, tonal balance, dynamics, stereo image and clipping. The result is shown in the **Song check** tab.



Guide value and notes for the loaded song

Guide value and notes

- The **number in the ring** (up to 100) is a guide value from measurements, not a judgement of the music. Green from 85, yellow from 65, otherwise red.
- Every **note** has a category, for example *Tonal balance*, *Stereo* or *AI artefacts*. Tonal notes are a matter of taste. Technical problems (such as clipping) weigh more.
- Notes with a button (for example **Lower bass**, **Bass mono on**) set up the matching module. **Apply all suggestions** does this for all of them at once.

Finding and smoothing AI artefacts



Spots found, each with a Listen button, below them the controls for smoothing

The card looks for spots typical of AI music: glitter and chirping in the highs, a fluttering stereo image and clicks. Spots found are marked violet in the waveform and listed with a **Listen** button.

1. **Listen to the spots**
   Tap **Listen** in the list. The studio jumps there and plays.

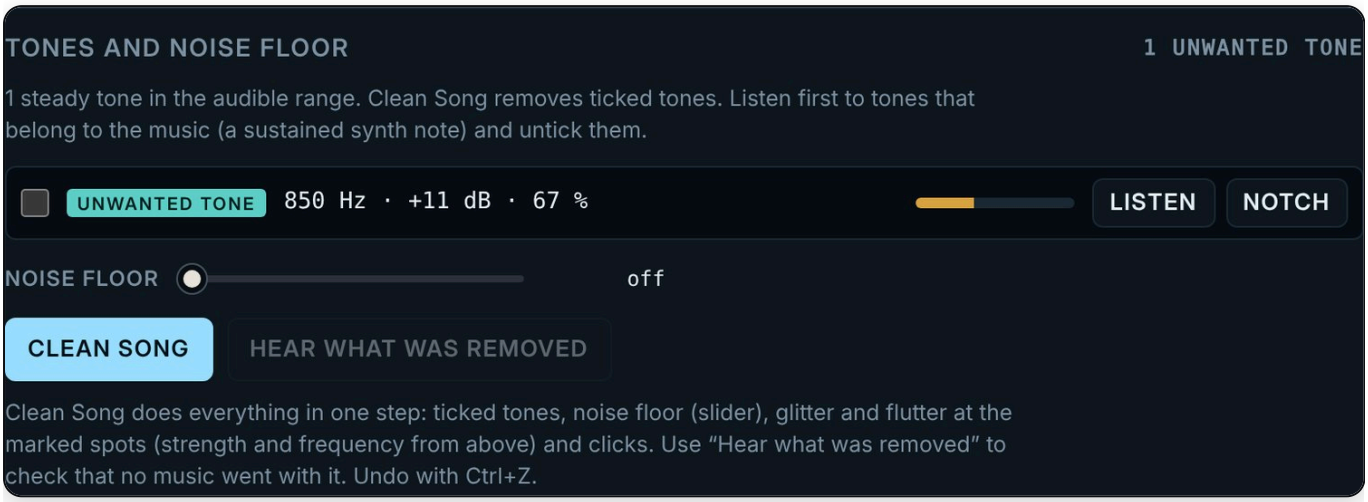
2. **Adjust**
   Strength sets how strongly to smooth. **From frequency** sets from which height upwards the tool works. Below that everything stays unchanged.

3. **Choose the range**
   Suspicious spots only, the current selection or the whole song. **Repair clicks** on or off.

4. **Smooth**
   Changes the song itself. With `Ctrl`+`Z` you hear the version before, with `Ctrl`+`Y` the smoothed one again.

Clean Song: remove unwanted tones and the noise floor

Some AI songs have a quiet whistle or grinding tone that sits at the same frequency throughout the song, or a constant noise in the highs. Smoothing above does not catch that because it only looks for short outliers. For this there is the **Tones and noise floor** section below the AI artefacts card.



Unwanted tones found, with tick boxes, the noise floor, the slider and the Clean Song button

1. **Read the list**
   The search runs on its own when the song loads. Each tone is listed with its frequency, how far it stands above the neighbouring frequencies (for example +28 dB) and in what percentage of the song it is present. A noise floor found is listed below with its frequency range.

2. **Listen and choose**
   Listen plays 6 seconds from the cursor of only this tone or range, made louder. If it sounds like music (a sustained synth note), untick it. Reliable finds are already ticked. Most people cannot hear tones above 18 kHz, but they disturb the limiter and MP3/OGG.

3. **Set the noise floor**
   The slider sets how strongly the constant noise is lowered. If a floor was found it is set to 60 %, otherwise off. Loud passages and attacks stay untouched.

4. **Clean Song**
   Removes in one step the ticked tones, the noise floor, glitter and flutter at the marked spots (strength and frequency from the card above) and the clicks. For a whole song this takes 10 to 30 seconds depending on the device.

5. **Check**
   Hear what was removed plays 8 seconds from the cursor of only what was removed (12 dB louder). If you hear music there, go back with `Ctrl`+`Z`, lower the slider or untick tones and try again.

6. **Or as a notch**
   The **Notch** button next to a tone adds it as a fixed, very narrow cut in the equalizer (chapter 6). It then acts on the master instead of the file. Untick the tone in the list if you do not want to remove it twice.

Voice: smooth the hiss (voice de-hisser)

AI vocals (for example from Suno) often carry a fine hiss or noise in the background that keeps the voice from sounding "round". The voice de-hisser below Clean Song works only where the voice sits: in the middle of the stereo image, only between 5 and 10 kHz and only while someone is singing. There it gently

lowers the noise between the harmonics of the voice. The harmonics themselves, the side, pauses, instrumental parts and everything below 5 kHz stay unchanged.



Strength, range, Apply, Before / After and Hear what was removed, with the report below

1. **Start low**
   Strength is set to 25 %. Go above 50 % only if you really prefer it in the comparison. At 100 % the reduction is at most 9 dB.

2. **Apply**
   For the whole song or only the selection. The report below shows how long singing was detected. If no singing is detected, the song stays unchanged.

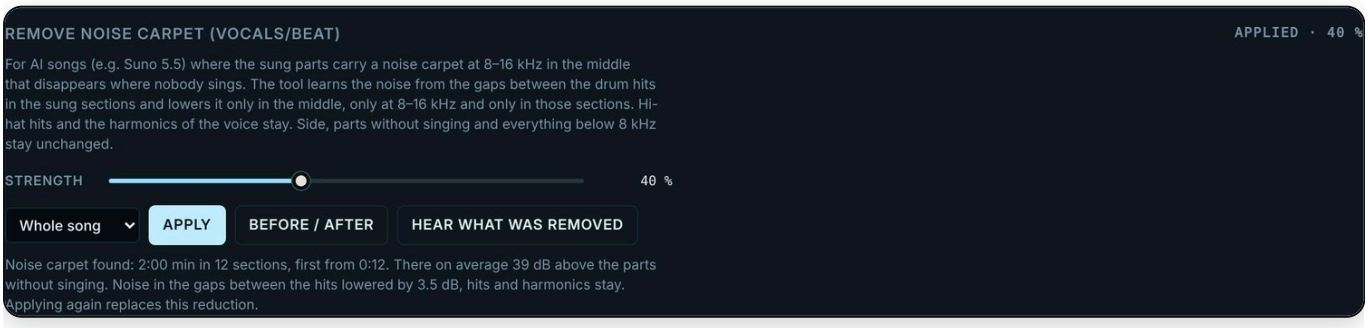
3. **Compare**
   Before / After switches during playback without interrupting. **Hear what was removed** plays 8 seconds from the cursor of only what was taken away, made louder. It should sound like quiet noise.

4. **Too much?**
   If the voice sounds thinner, hollow or metallic, lower the strength and press **Apply** again: this replaces the previous de-hissing. `Ctrl` + `Z` removes it completely.

> **Instead of turning Clean up**
> The noise floor of Clean Song works on the whole middle, including the voice. For hiss in the voice the de-hisser is the gentler tool. That is why Clean “light” in the album tab and in the chat tool now lowers the noise floor only in the side.

Remove noise carpet (vocals/beat)

Some AI songs (for example from Suno 5.5) carry a noise carpet in the middle at roughly 8–16 kHz in the sung parts. It is only there when vocals and beat play together; where nobody sings it is 30–40 dB quieter. Clean Song does not find it because it estimates its noise profile from quiet spots, and the voice de-hisser only reaches 10 kHz. This tool learns the noise from the gaps between the drum hits in the sung sections and lowers it only there: only in the middle, only at 8–16 kHz, only in those sections. Hi-hat hits and the harmonics of the voice stay; the side, parts without singing and everything below 8 kHz stay unchanged.



Strength, range, Apply, Before / After and Hear what was removed, with the report below

1. **Start with 30–50 %**
   Strength is set to 40 %. At 100 % the carpet is lowered by at most 30 dB.

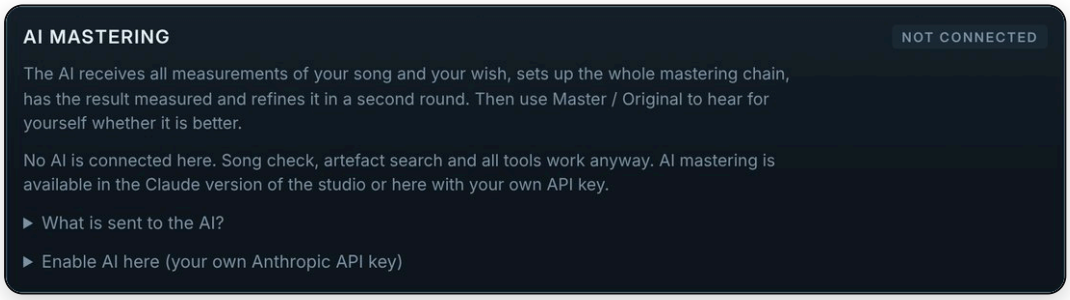
2. **Apply**
   For the whole song or only the selection. The report shows where a carpet was found, how far it lay above the parts without singing and how much the noise in the gaps is lowered now. Without a carpet the song stays unchanged.

3. **Compare**
   Ideally at a spot with vocals and beat: **Before / After** switches without interrupting. **Hear what was removed** plays 8 seconds from the cursor of only what was taken away, made louder. It should sound like even noise.

4. **Too much?**
   If the highs sound dull, “pumping” or full of holes, or you clearly hear hi-hats or words in what was removed, lower the strength and press **Apply** again: this replaces the previous reduction. `Ctrl` + `Z` removes it completely.

> **Together with the voice de-hisser**
> The two tools complement each other: the de-hisser for the hiss of the voice up to 10 kHz, the noise-carpet remover for the carpet above it. In the chat tool and the MCP server the setting is called **rauschteppich** (0–100 %, also per song).

AI mastering



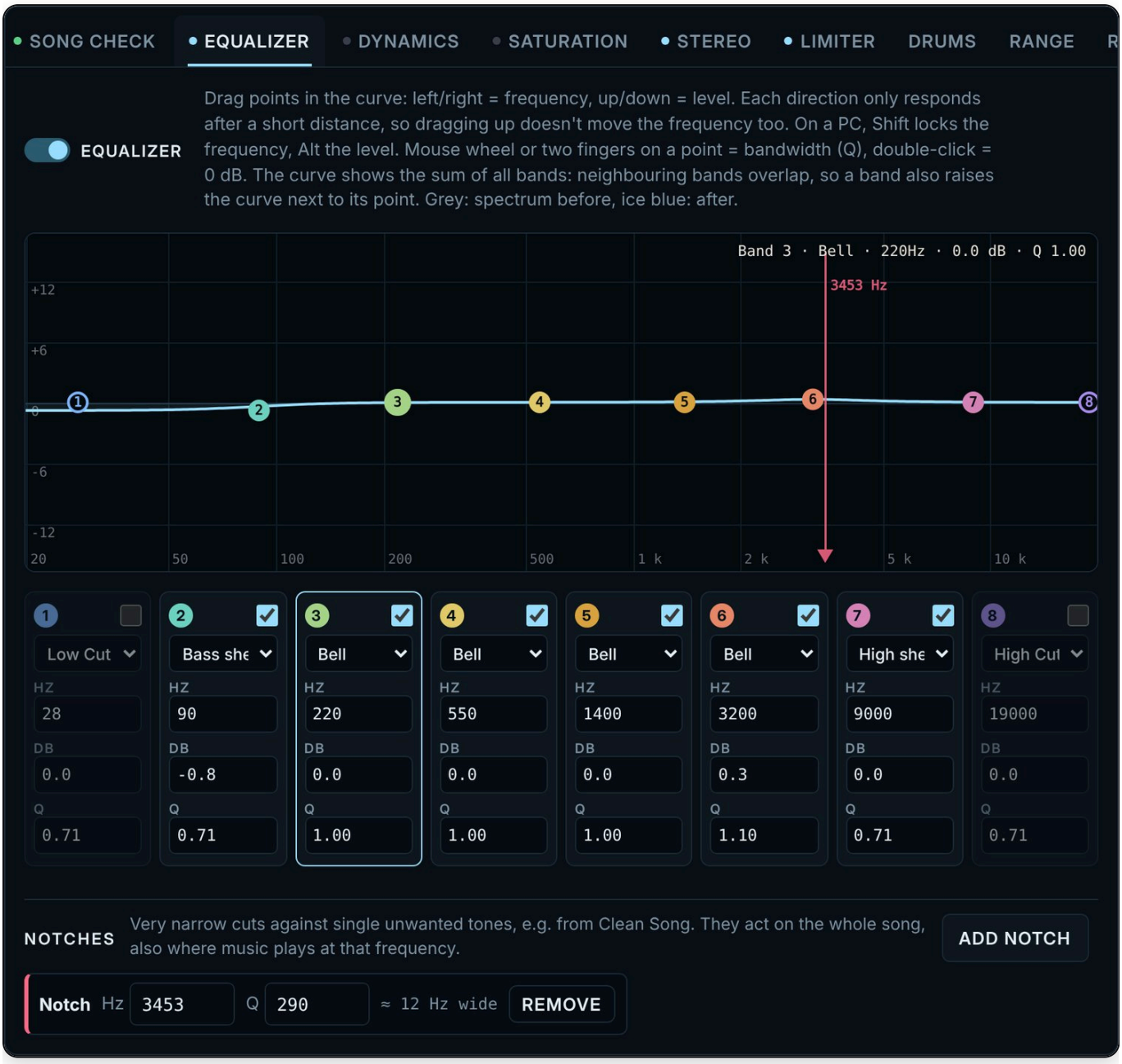
Describe in one sentence how your song should sound (for example “dark and punchy, bass up front, clear vocals, for Spotify”) and press **Master with AI**. The AI receives the measurements, sets up the whole chain, has the result measured and refines it in a second round. **Previous settings** undoes it. With the English interface the AI explains its changes in English.

- **Only numbers and text are sent,** never audio and no file name. “What is sent to the AI?” shows the full list.
- **In the Claude version** it runs via your Claude account. **On the website and on the PC** it needs your own API key from console.anthropic.com (a few cents per run). Without a key all other tools work as usual.

> **API key security**
> The key goes directly from your browser to Anthropic and by default is kept only in memory. Use a key just for this studio, set a low spending limit in the Anthropic console and do not enter it on other people's devices. Only tick “Remember on this device” on your own device.

6 Equalizer

The equalizer shapes the tonal balance: more or less bass, mids or highs. It has 8 bands. In the curve you see the spectrum before in grey and after in ice blue.



Curve with the 8 points and a notch at 3453 Hz (red mark), below it the bands and the notches

How to use it

- **Drag a point:** left/right = frequency, up/down = level
- **Mouse wheel** or **two fingers** on a point = bandwidth (Q). Small = wide, large = narrow.
- **Double-click a point:** level back to 0 dB
- Below the curve each band has a box to switch it on and off, the filter type and number fields.
- The **Equalizer** switch at the top left switches the whole module.

Filter types

Low Cut	removes low rumble below the frequency
Bass shelf	raises or lowers everything below
Bell	raises or lowers a range around the frequency
High shelf	raises or lowers everything above
High Cut	removes everything above the frequency

Notches against single unwanted tones

Below the bands there are up to six **notches**: very narrow cuts that remove exactly one frequency, such as a whistle. **Add notch** creates one; you enter frequency and Q. The higher the Q, the narrower the cut; at 3.4 kHz and Q 290 it is only about 12 Hz wide. Each notch appears as a red mark in the curve. The easiest way is from Clean Song with the **Notch** button. Notches are kept by the assistant, presets and AI mastering.

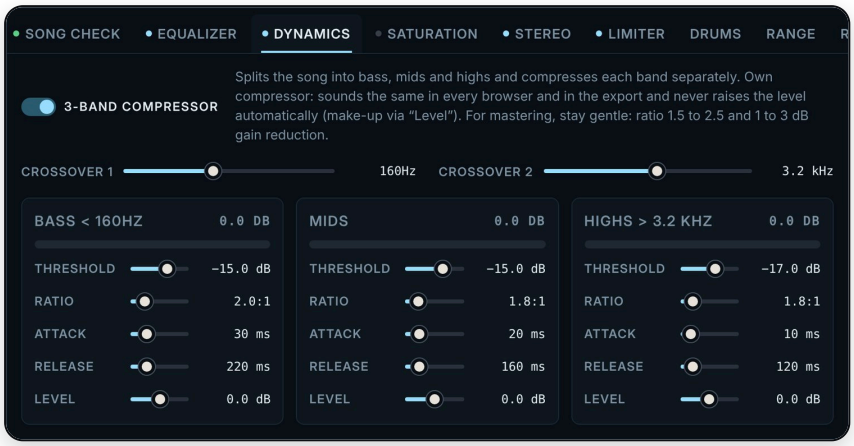
Notch or Clean Song? A notch always cuts, even when music is playing at that frequency. Clean Song only takes away the excess at each moment. For tones that are always present both work; for tones that change with the music, Clean Song is gentler.

> **Mastering means small steps**
>
> Changes of 1 to 3 dB are usually enough. Rather lower what bothers you than raise everything else. In the **Reference** tab the studio can calculate an EQ suggestion (chapter 10).

7 Dynamics, saturation, stereo

Dynamics (3-band compressor)

Splits the song into bass, mids and highs and compresses each band separately. This makes the song more even and denser. The compressor sounds the same in every browser and in the export and never raises the level automatically. You make up the level with **Level**.

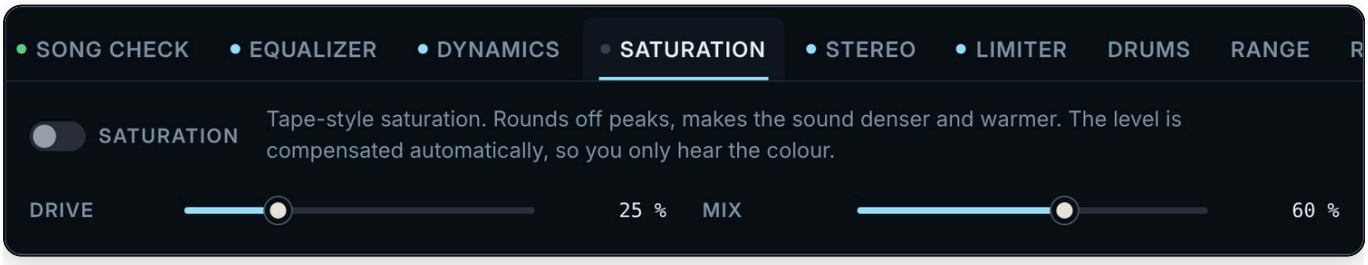


Crossover 1 / 2	Splits between bass and mids (default 160 Hz) and between mids and highs (3.2 kHz)
Threshold	Compression starts above this level. Lower = more effect.
Ratio	How strongly it compresses. 2:1 means: 2 dB above the threshold become 1 dB.
Attack / Release	How fast the compressor grabs and lets go again
Level	Volume of the band afterwards

The bar under each band shows how much is being reduced right now. For mastering stay gentle: ratio 1.5 to 2.5 and 1 to 3 dB of reduction.

Saturation

Tape-style saturation. Rounds off peaks, makes the sound denser and warmer. The level is compensated automatically, so you only hear the colour. **Drive** = strength, **Mix** = share of the signal.



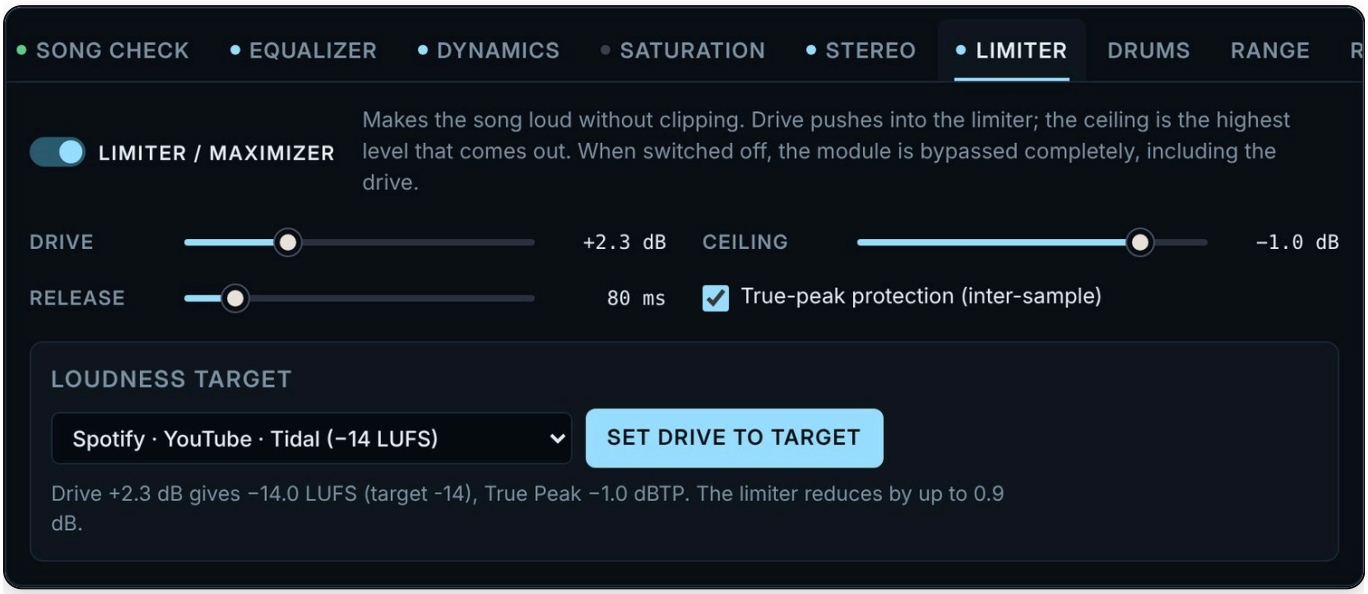
Stereo

Width above 100 % makes the song wider, below 100 % narrower. **Bass mono** puts everything below **Mono below** (default 120 Hz) in the centre. That gives punch and is almost a must for vinyl and the club.


SONG CHECK
EQUALIZER
DYNAMICS
SATURATION
STEREO
LIMITER
DRUMS
RANGE
R
STEREO
Width above 100 % makes the song wider, below 100 % narrower. Bass mono puts everything below the cut-off frequency in the centre. That gives punch and is almost a must for vinyl.
WIDTH
100 %
Bass mono
MONO BELOW
120Hz

8 Limiter and loudness

The limiter makes the song loud without clipping. It is the last module in the chain.



Drive	How hard the signal is pushed into the limiter. More = louder.
Ceiling	Highest level that comes out. For streaming, MP3 and OGG: −1 dB.
Release	How fast the limiter lets go. Short = louder, long = calmer.
True-peak protection	Also takes into account peaks between samples that arise during playback. Leave it on.
Switch off	The module is bypassed completely, including the drive.

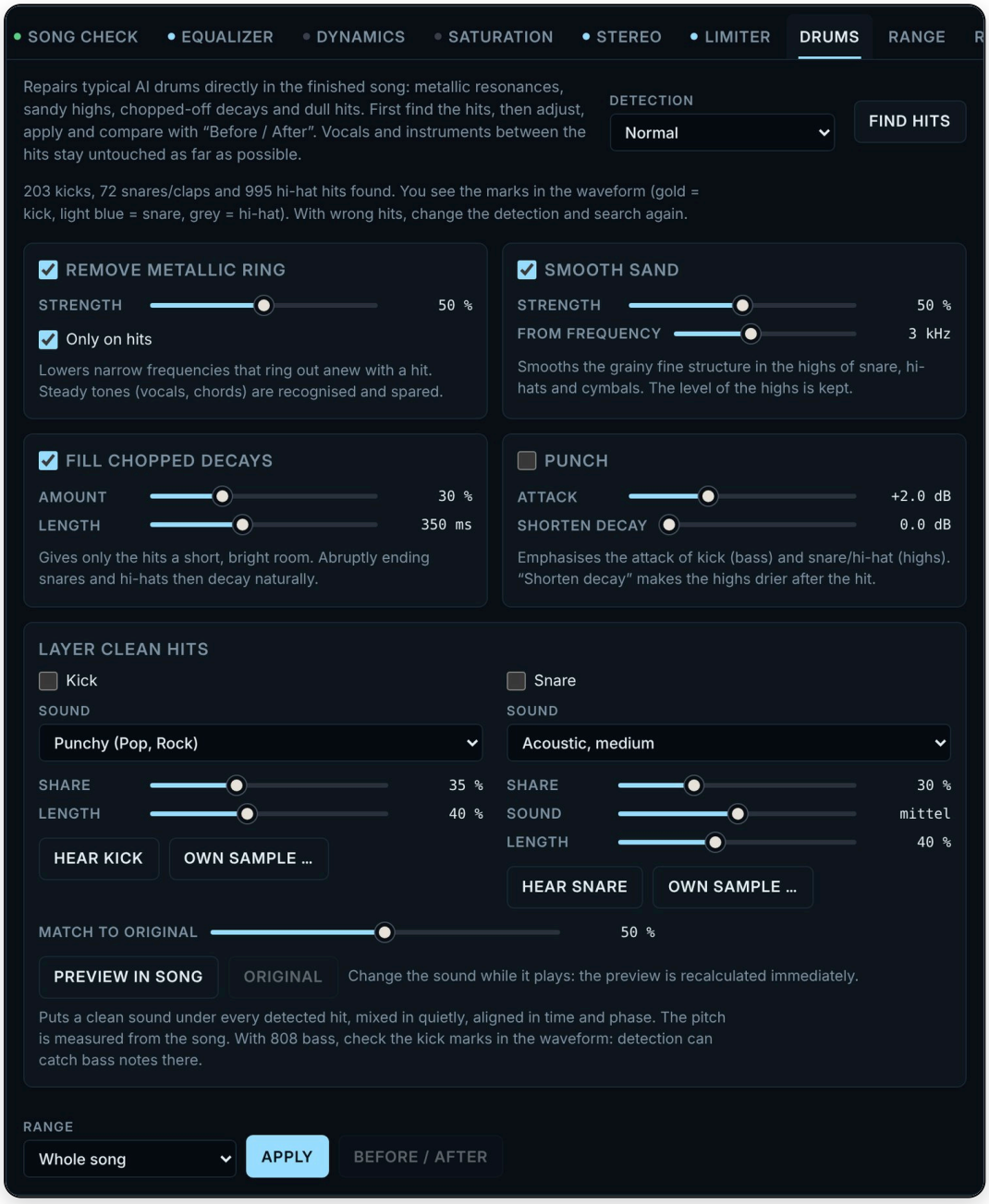
Loudness target in one step

1. **Choose a target**
   Spotify · YouTube · Tidal (−14 LUFS), Apple Music (−16), Loud, modern (−11), Club · SoundCloud (−9) or Vinyl premaster (−18, lots of headroom).
2. **Set drive to target**
   The studio calculates over the whole song how much drive is needed and sets it.
3. **Check**
   Below the button it says whether the target was reached and how hard the limiter works for it. Exact values are in the Analysis tab.

> **Louder does not help with streaming**
> Spotify, YouTube and others turn louder songs down to their level anyway. Louder than the target only costs dynamics there. If the limiter bar constantly shows more than 3 to 4 dB, it usually becomes audibly flat.

9 Drum repair

Repairs typical AI drums directly in the finished song: metallic resonances, sandy highs, chopped-off decays and dull hits. Vocals and instruments between the hits stay untouched as far as possible. Work with the finished WAV, not with stems.



Drums tab after "Find hits"

Step by step

1. **Find hits**
   Choose the detection (Conservative, Normal, Sensitive) and press **Find hits**. Afterwards it shows how many kicks, snares and hi-hats were found. You see the marks in the waveform. With wrong hits, change the detection and search again.

2. **Choose repairs**
   Each card has a tick box and controls:

- **Remove metallic ring**: lowers narrow frequencies that ring out with a hit. “Only on hits” spares the rest.
- **Smooth sand**: smooths the grainy fine structure in the highs above the set frequency. The level of the highs is kept.
- **Fill chopped decays**: gives the hits a short, bright room so they decay naturally.
- **Punch**: emphasises the attack. “Shorten decay” makes the highs drier after the hit.

3 **Layer clean hits (optional)**
For very bad drums: tick **Kick** and/or **Snare** and choose a sound. Kicks: Punchy, Deep and soft, 808 long, Dry and short, Acoustic, Hard with click. Snares: Acoustic medium, Acoustic dark and fat, Bright and snappy, Clap, Snare + Clap, Trap snare, Rimshot. **Hear kick** / **Hear snare** plays the sound on its own, **Own sample …** loads your own.

4 **Share and matching**
Share sets how loud the new sound is mixed in. **Match to original** adapts the timbre to the drums in the song (0 % = unchanged, 100 % = as close to the original as possible).

5 **Preview in song**
Plays 8 seconds from the playback position (or the selection, max. 20 s) with all settings in a loop. You can change sound and controls while it plays. **Original** switches for comparison. The song itself stays unchanged.

6 **Apply**
Choose the range (whole song or selection only) and **Apply**. Applying again replaces the previous repair instead of stacking it.

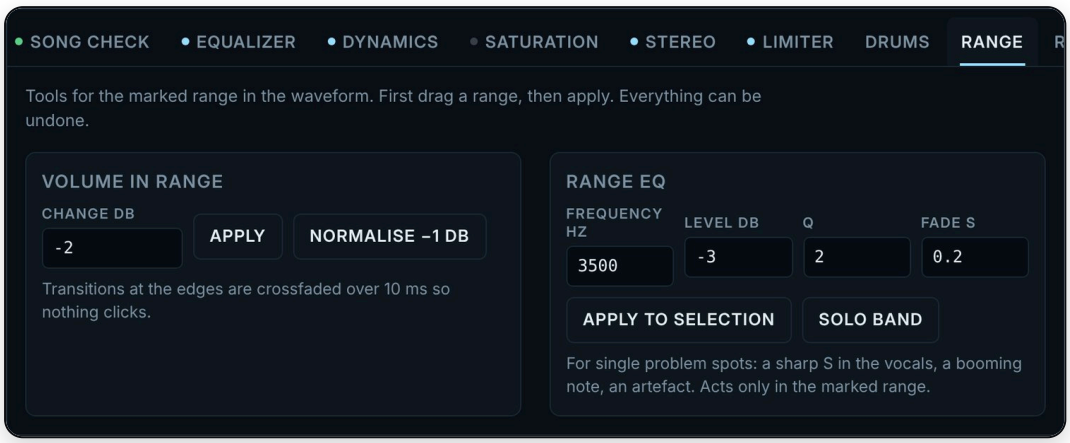
7 **Before / After**
Switches between original and repaired version without interruption. If you don't like it: `Ctrl` + `Z`.

> **Songs with 808 bass**
> Detection can mistake bass notes for kicks there. Check the gold kick marks in the waveform before layering kicks.

10 Range, reference, analysis

Range: tools for a single spot

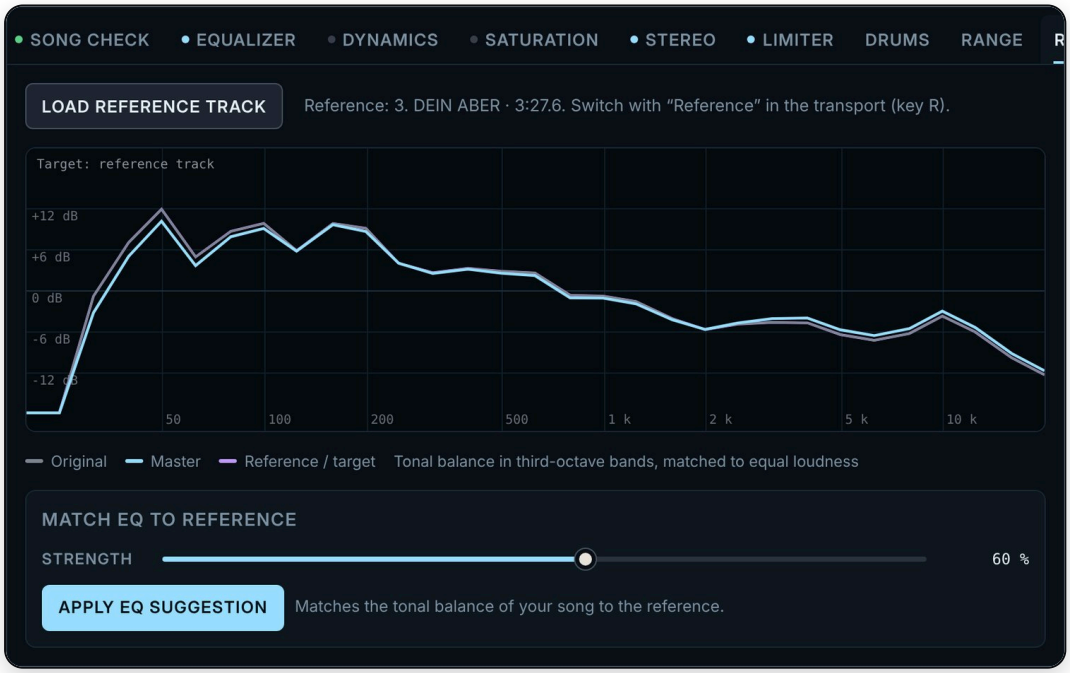
First mark a range in the waveform, then apply here. Everything can be undone.



- **Volume in range:** enter the change in dB and **Apply**. The edges are crossfaded over 10 ms so nothing clicks. **Normalise −1 dB** brings the loudest point of the selection to −1 dBFS.
- **Range EQ:** for single problem spots like a sharp S in the vocals, a booming note or an artefact. Enter frequency, level, Q and fade time, then **Apply to selection**. **Solo band** plays only this frequency band so you can find the spot.

Reference: compare with a model

Load reference track and choose a finished, mastered song that sounds the way you want. It plays in sync (monitoring: **Reference**) and is not processed.



Tonal balance of original, master and reference, matched to equal loudness

Match EQ to reference: choose a **Strength** and **Apply EQ suggestion**. The studio sets the equalizer so that the curve of your song moves closer to the reference. Without a reference it matches a balanced average curve. Then compare level-matched by ear.

Analysis: the exact values

Measured over the whole song according to ITU-R BS.1770, the same standard as Spotify and YouTube. Master values are re-measured automatically after every change. **Copy report** puts all values and notes on the clipboard.



Integrated loudness	Loudness of the whole song in LUFS. This is the value streaming services compare.
Short-term max.	Loudest 3-second spot
Loudness range (LRA)	How much the loudness varies within the song. Little = even, a lot = dynamic.
True Peak	Highest peak including inter-sample values. Should be below −1 dBTP.
Peak-to-loudness ratio (PLR)	How much headroom the peaks have above the loudness. Very small = heavily squashed.

Correlation, side to mid, stereo in the bass	Stereo image. A correlation below 0 points to phase problems.
Limiter max.	Largest reduction by the limiter

The curve below shows the short-term loudness over the length of the song, original and master on top of each other.

Listening comparison: what you actually hear

At the bottom of the Analysis tab the **Listening comparison** compares two versions through a model of hearing. What the music masks does not count. Loudness is matched first, so louder does not automatically count as different. The method is an approximation of the PEAQ and ViSQOL measurement standards, but not a certified standard value.



Result: original file against master

1. **What to compare?**
   Original file ↔ *Master* shows everything together. *Original file* ↔ *edit* checks only your repairs, for example Clean Song. *Edit* ↔ *Master* checks only the mastering.

2. **Compare**
   For a whole song this takes a few seconds.

3. **Read the number**
   5 = no audible difference, 4 = audible but not annoying, 3 = slightly annoying, 2 = annoying, 1 = very annoying. With repairs and mastering an audible difference is intended; the number says how clear it is, not whether it sounds better.

4. **Timbre**
   The six boxes show fixed EQ differences (bass to air). They are reported separately and not counted as a disturbance.

5. **Map and spots**
   The map shows when (to the right) and where in the frequency range (upwards) the difference is audible: yellow just audible, orange clear, red strong. Tap to play from that spot. Below are the most noticeable spots with **Listen**; switch between Original and Master at the top while listening.

When exporting as MP3 or OGG the comparison runs automatically between the master and the finished file and appears as “Audible encoding loss”. MP3 at 320 kbit/s reaches about 4.9, at 128 kbit/s about 4.5.

11 Game data

For games that run to the music: the studio delivers tempo, bars, song sections, energy and hits as data (JSON). All times in seconds from the start of the file.



1 **Open the tab**
The analysis starts on its own. **Analyse again** repeats it.

2 **Check the tempo**
Play and switch on **Click**: there is a click on every beat, a high tone on the 1. If the tempo is double or half: **÷ 2** or **× 2**. If the 1 is in the wrong place: **‹ 1** or **1 ›**. If you know the tempo: enter it under "Know the BPM?" and **Re-align grid**.

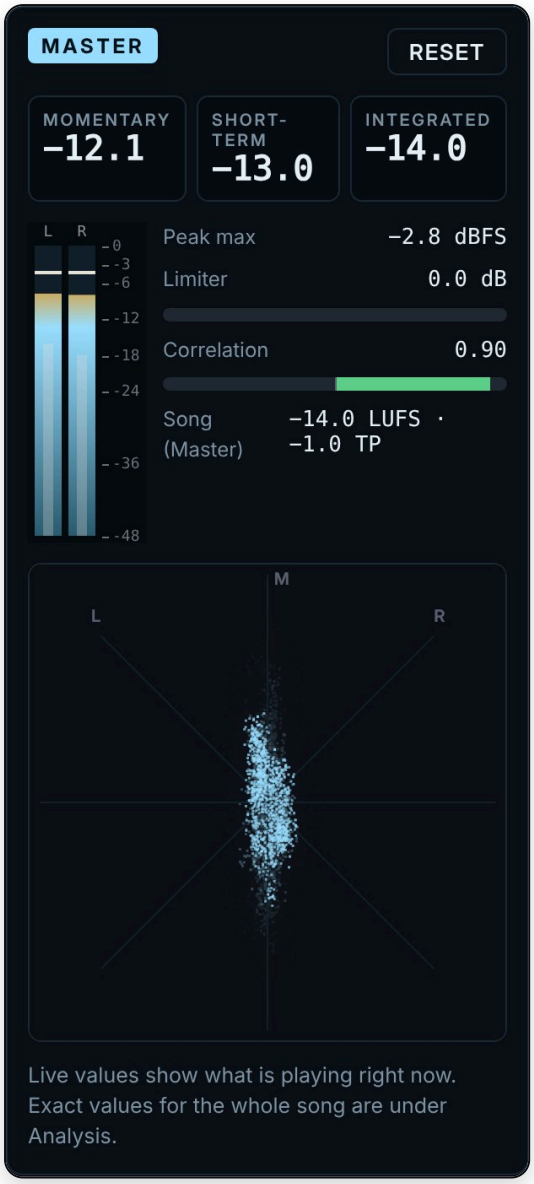
3 **Song details**
Enter title and level number. Under "Sections" choose coarse or fine division.

4 **Name the sections**
In the table give each section a type (for example Intro, Build-up, Drop) and a name for the game. Tapping the graphic plays from that spot.

5 **Choose content**
Tick boxes for beats, energy, bass/mids/highs, hits, loudness curve, grid and pretty-printing. The slider "Hits: only from strength" filters out quiet hits.

6 **Take it over**
Copy JSON or **Save as file**. **Copy Godot code** delivers a ready GDScript for loading. **Copy Codex brief** explains all fields for ChatGPT Codex. Section types in the JSON are English codes (intro, build, main …); sections you leave unnamed get the German default name (e.g. "Aufbau") as label, so name them yourself for an English game.

12 Meters



The meters on the right show what is playing right now. At the top it says which source you hear (Master, Original, Original file, Without mastering or Reference).

Momentary	Loudness of the last 0.4 s
Short-term	Loudness of the last 3 s
Integrated	Loudness since start or since **Reset**
L / R	Level left and right
Peak max	highest peak since reset
Limiter	how much the limiter is reducing right now
Correlation	+1 = mono, 0 = very wide, below 0 = phase problems
Song (Master)	measured loudness and true peak of the whole song
Vectorscope	the stereo image as a cloud. Narrow and vertical = rather mono, wide = lots of stereo.

Live values fluctuate with the music. The exact values for the whole song are in the Analysis tab.

13 Exporting and saving



1. **Choose content**
   Mastered (with all modules) or *Edit only* (your edited version without mastering).

2. **Choose format**
   see the table below.

3. **Sample rate**
   Same as original, 44.1 kHz or 48 kHz (video). MP3 only supports 32, 44.1 and 48 kHz. Other rates are converted to 48 kHz, with a note.

4. **Enter a file name and start the export**
   The studio renders the whole song, measures it and hands over the file. It is then in your downloads folder. On a phone, choose “Save to Files” in the share menu.

5. **Read the result**
   Below the file name you see loudness, true peak and dynamics. For MP3 and OGG the finished file is measured again; below that are the values before encoding and the audible encoding loss (1 to 5, chapter 10). It is normal that MP3 and OGG change the peaks slightly. If the true peak of the finished file is above 0 dBTP, the studio warns you: then lower the limiter ceiling a little and export again.

Format	What for
WAV 24 bit	Distribution (distributor, streaming), archive. The best choice for releases.
WAV 16 bit	CD, with dither
WAV 32 bit float	Further processing in another program
OGG Vorbis ~140 / ~240 kbit/s	Games and web. Gapless, ideal for music in Godot.
MP3 320 / 192 kbit/s	For sharing and listening. 320 = best MP3 quality, 192 = smaller.

A mastered export is also remembered as a draft for My Style (chapter 14). It only counts once you confirm it there.

Saving settings

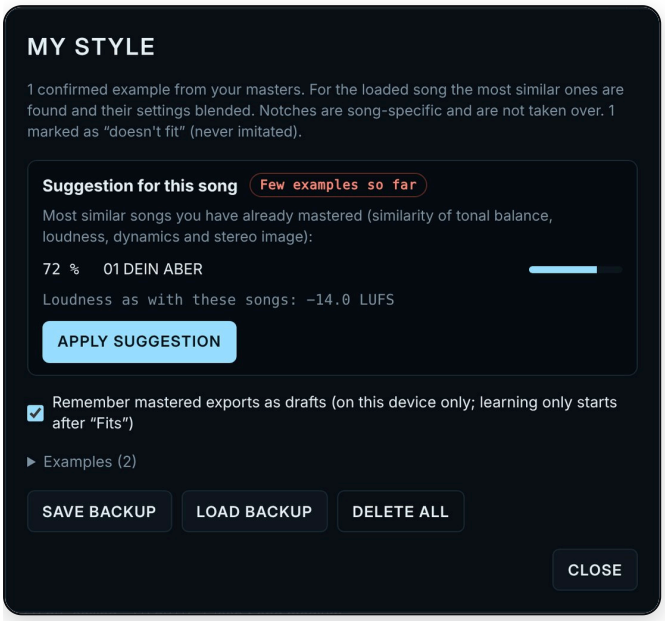
Save settings stores all controls as a file, **Load** brings them back. This way you can keep your own settings for every album or song and later continue exactly where you left off.

Feedback

Feedback opens a form with stars, praise, criticism and optionally name and genre. **Send by e-mail** opens your e-mail program with a ready text, **Copy text** e.g. for WhatsApp. If you like, measurements and settings are attached, never audio.

14 My Style

My Style learns from your own masters. For a new song it looks for the most similar songs you have already mastered and confirmed, and suggests settings in your style.



Suggestion with the most similar songs and the examples

How examples are created

- **Export:** every mastered export is remembered as a *draft* (on this device only). A draft does not count yet.
- **Confirm after listening:** in My Style, under **Examples**, tap **Fits**. Only then does the studio learn from it. **Doesn't fit** marks an example that is never imitated.
- **Album:** an approved song (chapter 15) automatically becomes a confirmed example.

This way the studio only learns from masters you have really listened to and found good, and never from mistakes.

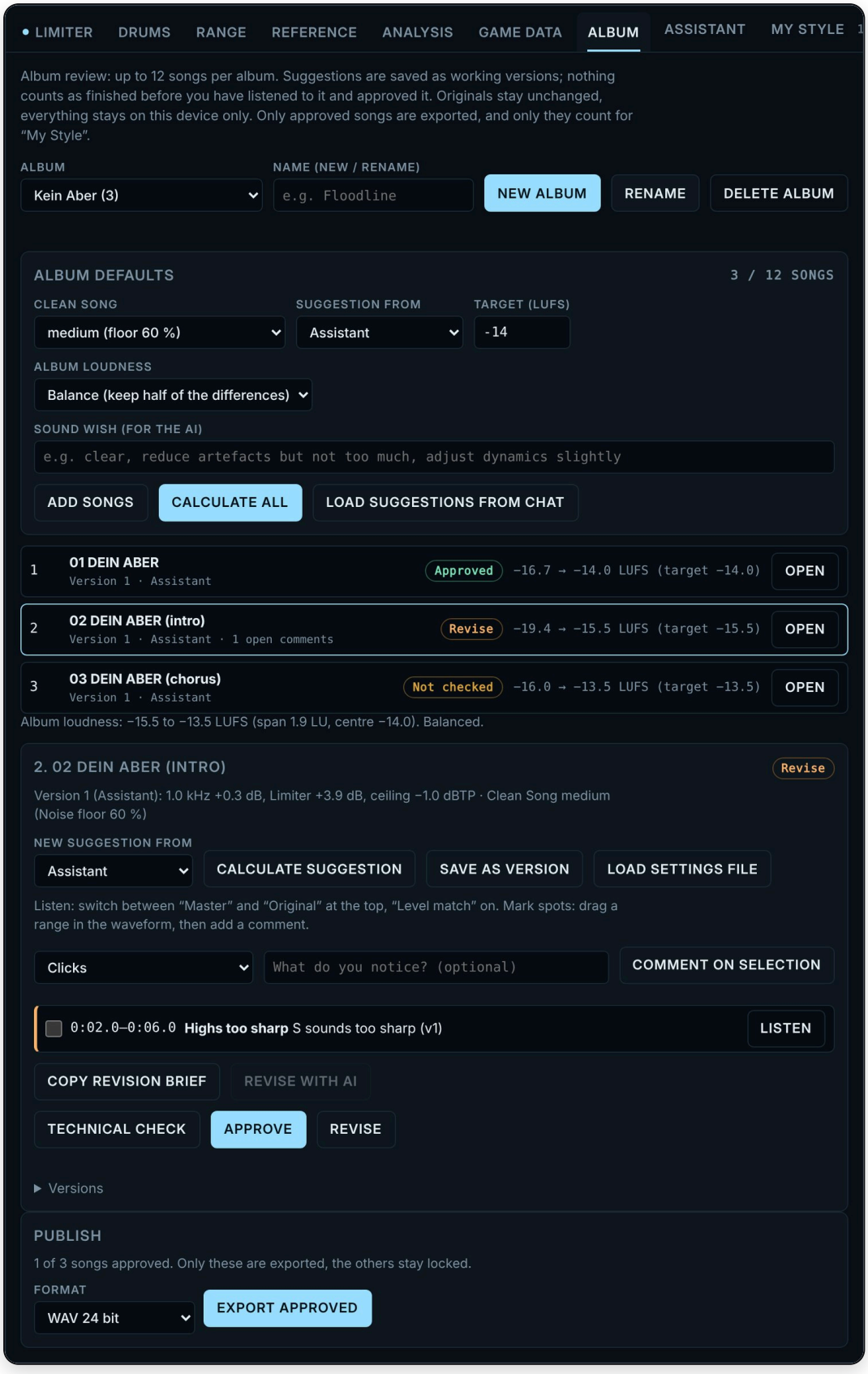
Applying a suggestion

1. **Load a song**
   and wait until it has been measured.
2. **My Style**
   Press it next to the assistant. The number on the button shows how many confirmed examples there are.
3. **Read the suggestion**
   The most similar songs are listed with their similarity in percent (tonal balance, loudness, dynamics, stereo image). The label says how reliable the suggestion is: *Few examples so far*, *Suggestion usable* or *Suggestion reliable*.
4. **Apply suggestion**
   The studio blends the settings of the most similar songs and sets the loudness. Notches are not taken over, they belong to one particular song. Then fine-tune by ear.

From 20 to 30 examples on it gets good
From 15 confirmed examples on, the studio also learns which measurements matter most for your style. **Save backup** puts all examples in a file, **Load backup** brings them e.g. to your phone.

15 Album: review and approve

In the **Album** tab you work on up to 12 songs as an album. Suggestions are working versions. A song is only finished once you have listened to it and approved it. The originals stay unchanged, everything stays on this device only.



Album with three songs: approved, marked for revision, not yet checked

1. **Create an album**
   Enter a name and press **New album**. **Rename** and **Delete album** only affect the studio; your files stay untouched.

2. **Choose defaults**
   Clean Song (off, light, medium, strong), where the suggestions come from (Assistant, My Style, AI or Neutral), loudness target, album loudness (*Balance* keeps half of the differences between the songs, or all equally loud) and a sound wish for the AI.

3. **Add songs**
   Up to 12 audio files; they are sorted by name.

4. **Calculate suggestions**
   Calculate all creates a suggestion for every song without a version. After a stop or restart it continues there. If you have worked in the “Mastering” chat: **Load suggestions from chat** imports the *.dsmaster.json* files, matched by name.

5. **Listen and comment**
   Load a song with **Open**, switch between Master and Original at the top (level match on). For a spot, drag a range in the waveform, choose the issue type and press **Comment on selection**. Comments appear as marks in the waveform.

6. **Make changes**
   Keep your own changes with **Save as version**, calculate a new suggestion or load a settings file. Every version can be reopened under **Versions**.

7. **Check and decide**
   Technical check, then **Approve** or **Revise**. For a revision: **Copy revision brief** and paste it into the chat, or **Revise with AI**.

8. **Publish**
   Choose the format and **Export approved**. On a PC you choose a folder; “Album – Master” is created inside it with the files and a mastering log.

Status	Meaning
Not checked	Suggestion available, not yet listened to and decided
Revise	You marked the song for revision; comments are open
Approved	Listened to, technically checked, approved. Only these songs are exported and count for My Style.
Technically blocked	The check found an error that prevents approval

What the technical check looks at

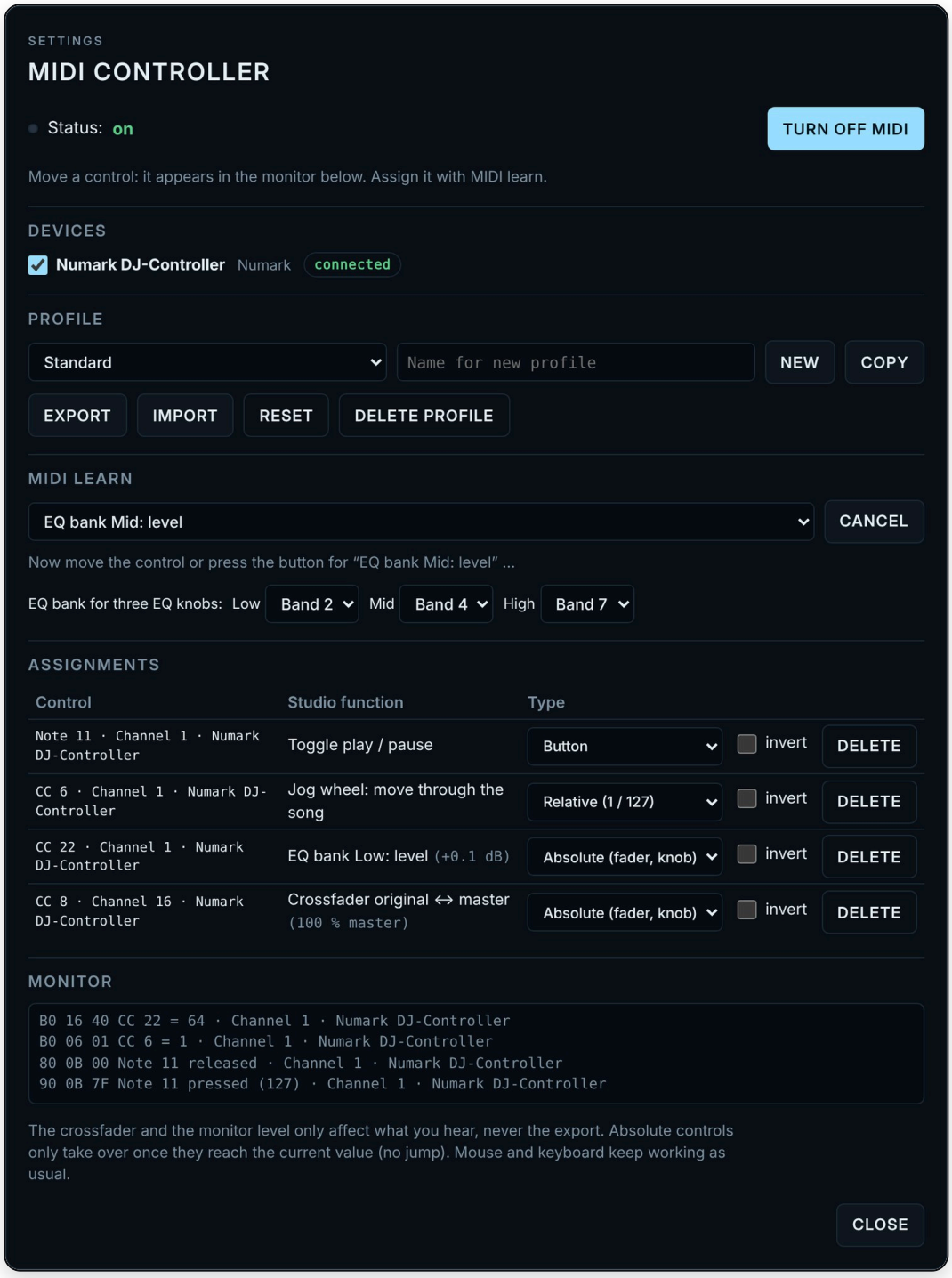
Invalid samples, clipping, true peak above the ceiling, length compared with the original, a silent channel, phase and bass in mono. Notes without blocking: DC offset, deviation from the loudness target and a strong change in sound in the listening comparison. Below the song list the *Album loudness* line shows how far apart the songs are.

Approve means: listened to and found good

If an approved song changes afterwards, it no longer counts until it is approved again, neither for the export nor for My Style.

16 MIDI controller

From version 1.15.0 the studio can be controlled with a DJ or MIDI controller: assign knobs, faders, the jog wheel and pads freely, with a crossfader for comparing original ↔ master. Mouse and keyboard keep working as usual.



Settings → MIDI controller: device, profile, MIDI learn, assignments and monitor

Connect the controller

1. **Connect**
   Plug the controller in via USB. **Close Serato or any other DJ program first**: on Windows only one program can use the device at a time.

2 **Turn on**
Click **MIDI** at the top, then **Turn on MIDI**. The browser asks once whether the page may use MIDI devices: **Allow**.

3 **Check**
Under **Devices** the controller shows as “connected”. When you move a control, the dot next to “MIDI” lights up and the message appears in the **Monitor**.

MIDI learn: assign controls

1 **Choose a function**
In the field under MIDI learn, for example “EQ bank Low: level” or “Jog wheel: move through the song”.

2 **Learn**
Press **MIDI learn**, then briefly move the control or press the button on the controller. Done.

3 **Adjust**
In the **Assignments** table you can change the type (absolute, relative, button), invert the direction or delete the assignment.

- **No jumps:** a fader or knob only takes over once it reaches the current value. If it is elsewhere, a message tells you which way to move it.
- **EQ bank:** the three EQ knobs of a DJ controller control three freely chosen bands out of eight (default: bands 2, 4 and 7).
- **Crossfader and monitor level** only affect what you hear, never the export.
- **Profiles** store the assignments per controller; export and import for another computer.

> **Privacy**
> The studio only receives control data, sends nothing to devices and does not use SysEx. The full list of assignable functions is in docs/MIDI.md.

17 Keys and gestures

Keyboard

`Space`	Play / pause
`L`	Loop on / off
`M` `O` `R`	Hear master / original / reference
`+` `−`	zoom in / zoom out
`Home`	to the start of the song
`Esc`	clear selection, close window
`Ctrl`+`A`	select the whole song
`Ctrl`+`X` / `C` / `V`	cut / copy / paste
`Del`	delete selection
`Ctrl`+`Z`	undo
`Ctrl`+`Y`	redo (also `Ctrl`+`Shift`+`Z`)

On a Mac use `⌘` instead of `Ctrl`.

Mouse

Click	set the playback position
Drag	mark a range
Shift + click	extend the selection
Mouse wheel	zoom (in the waveform), bandwidth (in the EQ)
Shift + mouse wheel	scroll sideways
Double-click waveform	show the whole song
Double-click slider	default value
Double-click EQ point	level to 0 dB

Touch (phone, tablet)

Tap	set the playback position
Drag	mark a range
Two fingers	zoom (waveform), bandwidth (EQ point)
Drag the timeline	move the playback position

18 Tips and help

Good habits

- **Always compare level-matched.** Louder almost always sounds better; that is deceptive.
- **Take breaks.** After 20 to 30 minutes your ears get used to it. Step away briefly, then listen to master against original again with fresh ears.
- **Check on several devices:** headphones, speakers, phone, car.
- **Less is more.** A good master changes the song only slightly. If you prefer the original, that is a clear sign.
- **One song, one settings file.** With “Save settings” you can continue at any time, and all songs of an album sound consistent.

Frequently asked questions

Problem	Solution
No sound	Tap or click once into the page (browsers only allow sound after an interaction). Check the device volume. With “Reference” but no reference loaded, the studio jumps to the Reference tab.
Master and original sound the same	Check whether modules are switched on (dot on the tab) or the “Neutral” preset is loaded. Above the meter panel it says what is playing.
It sounds distorted	Lower the drive in the limiter. A limiter bar above 4 dB means: pushed too hard.
Loudness target not reached	The studio reports this. Usually the ceiling is too low or the target very loud. Ceiling at −1 dB, target −14 LUFS.
File loads with a note	Incomplete files, multichannel files (for example 5.1) and unusual headers are loaded anyway. The note says what was adjusted.
MP3/OGG export takes a while	The encoder is loaded the first time. After that it also works offline. A whole song takes about 20 seconds to a minute depending on the device.
Some text is still German	Song and file names stay as they are. Files the studio creates for chats and games (revision brief, Codex brief, Godot code, mastering log) are in German. Everything else should be English: please report any German text via Feedback.
New version does not appear	Close the studio once and open it again. The version number is shown at the top left next to the logo.

> **Feedback**
> Something unclear, a bug, a wish? The **Feedback** button sends it straight to Deep Structure.